gwflow

Purpose

The gwflow module writes its own set of output files when `bsn_cc%gwflow = 1` in `codes.bsn`. Outputs are toggled in `print.prt` via six rows: `gwflow_wb`, `gwflow_flux`, `gwflow_heat`, `gwflow_solute`, `gwflow_obs`, `gwflow_pump`. Each row has the standard daily/monthly/yearly/average-annual flags.

Source

- Writer: `src/gwflow_output.f90`
- Types: `src/gwflow_module.f90`, `type groundwater_ss` for fluxes
- Simulation-time writes (cell-level): `src/gwflow_simulate.f90`

Output files

Water balance

File	Scope	Aggregation
`gwflow_basin_wb_day.txt`	basin total	daily
`gwflow_basin_wb_mon.txt`	basin total	monthly
`gwflow_basin_wb_yr.txt`	basin total	yearly
`gwflow_basin_wb_aa.txt`	basin total	annual average
`gwflow_cell_wb_day.txt`	per cell	daily
`gwflow_cell_wb_mon.txt`	per cell	monthly
`gwflow_cell_wb_yr.txt`	per cell	yearly
`gwflow_cell_wb_aa.txt`	per cell	annual average

Columns (in declaration order from `type groundwater_ss`, units `m^3`):

#	Name	Description
1	chng	change in storage (grid summaries only)
2	rech	recharge
3	gwet	groundwater ET
4	gwsw	groundwater discharge to channels
5	swgw	channel seepage to groundwater
6	satx	saturation excess flow
7	soil	groundwater added to the soil profile
8	latl	lateral flow between cells
9	disp	dispersion (heat and solute transport)
10	bndr	boundary exchange
11	ppag	allocation-driven pumping (irrigation)
12	ppdf	pumping deficit (unmet demand)
13	ppex	external pumping
14	tile	tile drainage outflow
15	resv	reservoir exchange
16	wetl	wetland exchange
17	fpln	floodplain exchange
18	canl	canal exchange

#	Name	Description
19	pond	recharge pond seepage
20	phyt	phreatophyte transpiration
21	totl	sum of inputs and outputs

Per-cell files prepend a cell id and timestamp; basin files aggregate across all cells. The full per-cell water-balance header (`gwflow_cell_wb_*`), verified against `gwflow_output.f90`, is:

```
jday mon day yr unit gis_id name head wt_depth recharge gw_et gw_sw sw_gw sat_excess soil lateral pump_all
```

- `gis_id` is the authoritative cell id carried from the QSWAT+ shapefile; `name` is `cellNNNN` (a 12-character field).
- `head` and `wt_depth` are in metres; the flux columns (`recharge` through `phytorem`) are in m^3/day for daily files (monthly files carry the monthly-average daily rate).
- `wt_depth` = cell land-surface elevation − head (depth of the water table below the cell's representative ground surface). It can look deeper at channel/river cells on coarse grids: the cell surface represents the valley, not the streambed, so the depth is measured from the valley rim while the head sits near the channel. This is a grid-resolution reporting effect, not a model error; the head field itself is correct.

Observation wells

`gwflow_obs_{day,mon,yr,aa}.txt` report state at each observation cell (set via the observation list) for every timestep. Columns:

Column	Units	Meaning
`head`	m	Groundwater head
`wt_depth`	m	Water-table depth below ground (land-surface elevation − head)
`temp`	degC	Groundwater temperature, `-99` when heat transport is off
`no3`	mg/L	Nitrate concentration, `-99` when solute transport is off
`p`	mg/L	Phosphorus concentration, `-99` when solute transport is off

Heat

When the heat transport sub-module is active (`gwflow.heat` is present), the following are written:

File	Aggregation
`gwflow_basin_heat_day.txt`	daily
`gwflow_basin_heat_yr.txt`	yearly
`gwflow_basin_heat_aa.txt`	annual average

Columns are the same fields as water balance, reinterpreted as heat fluxes (the writer reuses `type groundwater_ss`). Refer to the model theory page on Aquifers for the heat transport equations.

Solutes

When solute transport is active (`gwflow.solutes` listing constituents), per-species water-balance-style files are written. Six species are emitted directly by the writer:

Species	Files
Calcium (`ca`)	`gwflow_basin_sol_ca_{day,mon,yr,aa}.txt`
Chloride (`cl`)	`gwflow_basin_sol_cl_{day,mon,yr,aa}.txt`

Species	Files
Potassium (k)	gwflow_basin_sol_k_{day,mon,yr,aa}.txt
Magnesium (mg)	gwflow_basin_sol_mg_{day,mon,yr,aa}.txt
Sodium (na)	gwflow_basin_sol_na_{day,mon,yr,aa}.txt
Phosphorus (p)	gwflow_basin_sol_p_{day,mon,yr,aa}.txt

Columns follow the same `groundwater_ss` field set, interpreted as solute mass fluxes (kg).

Cell definition

`gwflow_cell_definition.txt` is written once at the start of the run. It maps the grid coordinates to cell IDs and is the key for interpreting any per-cell output.

Always-on log

`gwflow_record` is a plain-text log opened when `bsn_cc%gwflow = 1`. The simulation banner and per-feature notes are written there. Useful for confirming the gwflow module activated and which sub-features (pumps, ponds, tiles, etc.) loaded.

print.prt toggles

Toggle	Files produced
`gwflow_wb`	basin and cell water balance
`gwflow_flux`	gwflow canal, pond, tile, gwsw, channel observation diagnostics
`gwflow_heat`	basin heat balance
`gwflow_solute`	basin solute balance for each species
`gwflow_obs`	observation well output (when `gwflow.hru_pump_observe` or observation file is set)
`gwflow_pump`	per-HRU pumping output

The aggregation flags on each row (`d`, `m`, `y`, `a`) control which suffix variants are written.

Related pages

- `gwflow` input reference
- `codes.bsn` (`gwflow` switch)
- `print.prt` (output toggles)
- Model theory: Aquifers